

EE3621: Digital Signal Processing

Lecture 20: IIR Filter - Lattice Form Realization (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Convert** FIR direct form coefficients to lattice reflection coefficients k_m using the step-down algorithm.
 2. **Evaluate** IIR filter stability directly from reflection coefficients.
 3. **Construct** signal flow graphs for Lattice form realization.
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2. MATHEMATICAL FOUNDATIONS

2.1 FIR Lattice Recursion & Step-Down Algorithm

- **Lattice Recursion:**

$$A_m(z) = A_{m-1}(z) + k_m z^{-1} B_{m-1}(z)$$
$$B_m(z) = z^{-m} A_m(z^{-1}) \quad (\text{Reverse Polynomial})$$

- **Step-Down Algorithm (Direct $\rightarrow$ Lattice):**

Given $A_N(z) = 1 + \sum_{i=1}^N \alpha_N(i) z^{-i}$: 1. Set $k_N = \alpha_N(N)$. 2. For $m = N, N-1, \dots, 2$:

$$\alpha_{m-1}(i) = \frac{\alpha_m(i) - k_m \alpha_m(m-i)}{1 - k_m^2}, \quad i = 1, 2, \dots, m-1$$

$$k_{m-1} = \alpha_{m-1}(m-1)$$

Pictorial Representations

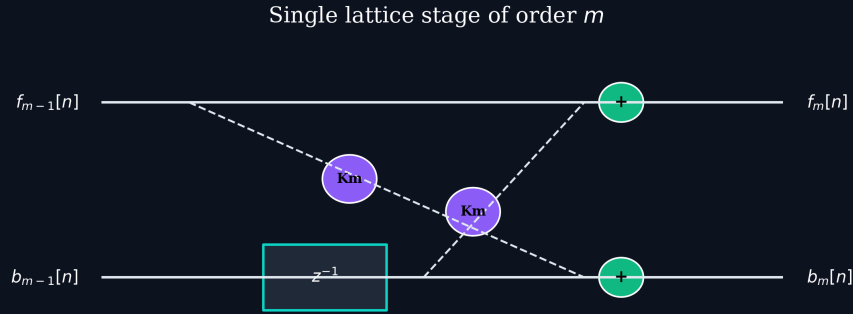


Figure 1: Fir Lattice Stage

3. WORKED NUMERICAL EXAMPLES

Example 20.1: Step-Down Conversion to Reflection Coefficients

Problem: Find the reflection coefficients k_1, k_2, k_3 for the FIR filter:

$$A_3(z) = 1 - 0.9z^{-1} + 0.64z^{-2} - 0.5z^{-3}$$

Solution:

- **Stage 3:**

$$k_3 = \alpha_3(3) = -0.5. \quad 1 - k_3^2 = 1 - (-0.5)^2 = 1 - 0.25 = 0.75.$$

- **Compute $\alpha_2(i)$ for $m = 3$:**

$$\begin{aligned} * \alpha_2(1) &= \frac{\alpha_3(1) - k_3 \alpha_3(2)}{0.75} = \frac{-0.9 - (-0.5)(0.64)}{0.75} = \frac{-0.9 + 0.32}{0.75} = \frac{-0.58}{0.75} = -0.7733. \\ * \alpha_2(2) &= \frac{\alpha_3(2) - k_3 \alpha_3(1)}{0.75} = \frac{0.64 - (-0.5)(-0.9)}{0.75} = \frac{0.64 - 0.45}{0.75} = \frac{0.19}{0.75} = 0.2533. \end{aligned}$$

- **Stage 2:**

$$1 - k_2^2 = 1 - (0.2533)^2 = 1 - 0.0642 = 0.9358.$$

- **Compute $\alpha_1(i)$ for $m = 2$:**

$$* \alpha_1(1) = \frac{\alpha_2(1) - k_2 \alpha_2(2)}{0.9358} = \frac{-0.7733(1 - 0.2533)}{0.9358} = \frac{-0.7733 \times 0.7467}{0.9358} = -0.6170. \quad k_1 = \alpha_1(1) = -0.6170.$$

- **Stability Check:**

$|k_1| = 0.6170 < 1$, $|k_2| = 0.2533 < 1$, $|k_3| = 0.5000 < 1$. All $|k_m| < 1 \implies$ The corresponding all-pole filter is **Stable**.